

Arithmetic restriction, source projections, and finite volume certificates

13 September 2026

This article joins the original arithmetic lower-envelope comparison to the canonical restriction maps. It proves a positive arithmetic return gap at every finite degree, carries entry errors through the actual quotient, and identifies all loss traces with observations of two original source projections. The balanced product gives a complementary lower estimate on every inverse spectral power in both endpoint windows. The complete arithmetic envelope and balanced monic-norm proofs are retained as explicit source dependencies; their constants appear below. Every finite matrix and restriction consequence used here is proved.

1 The original source, quotient, and canonical lifts

Retain $g = 2\xi$, the monic packet h of selected zeros with their complete orders, $v_h = g/h$, and

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad m_k = w_h^{*k}, \quad c = k/2. \quad (\text{ERJ.1})$$

The original pairing, conjugate-linear in its first entry, is $\langle P, Q \rangle = \int \overline{P(c + iu)} Q(c + iu) m_k(u) du$. Its constant norm is μ_h^k . The arithmetic construction supplies positivity almost everywhere and all polynomial moments. A nonzero polynomial has finitely many zeros on this line, so its norm is positive. Let $S = s_1 + \dots + s_k$, let the monic generator of $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$ be χ , and put $E = \mathbb{C}[S]/(\chi)$, $q = \deg \chi \geq 1$, $A = M_S$. The ordered quotient coordinates are $1, S, \dots, S^{q-1}$. For $N \geq q - 1$ write $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{D}_N = \chi \mathcal{P}_{N-q}$, with negative degree spaces zero. Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} E \longrightarrow 0. \quad (\text{ERJ.2})$$

In monomial coordinates let H_N be the source Gram. In the original monic orthogonal basis $p_0, \dots, p_N$, put $O_N = \text{diag}(\omega_0, \dots, \omega_N)$ and $B_N = [b_0, \dots, b_N]$, $b_n = [p_n]_{\chi}$. All p_n stay monic; $\omega_0 = \mu_h^k$. The monic triangular coefficient change has determinant one and gives the equal constructions

$$\begin{aligned} K_N &= J_N H_N^{-1} J_N^* = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad V_N = \det G_N, \\ R_N &= O_N^{-1} B_N^* G_N : (E, G_N) \longrightarrow (\mathcal{P}_N, O_N), \quad K_N = \sum_{n=0}^N \frac{b_n b_n^*}{\omega_n}. \end{aligned} \quad (\text{ERJ.3})$$

Here and below a formula with O_N uses monic source coordinates; its monomial version conjugates by that explicit coefficient change. Surjectivity of B_N proves $K_N > 0$. Multiplication gives $B_N R_N = I$ and $R_N^* O_N R_N = G_N$. For $z \in \ker B_N$, $z^* O_N R_N = z^* B_N^* G_N = 0$. Thus every lift $R_N x + z$ has squared norm $x^* G_N x + z^* O_N z$. This proves the unique minimum property, $\text{im } R_N = \mathcal{D}_N^\perp$, and, for $i \leq j$, $G_j \preceq G_i$ and $K_i \preceq K_j$.

The actual analytic representative is $\mathcal{V}P = P(D_1 + \dots + D_k) F_h^{\otimes k}$, where $D = -x\partial_x$, $\mathcal{M}F_h = v_h$, and $h(D)F_h = \Theta\phi_0$. Its norm is the pairing in (ERJ.1). Its full-jet map is $J^{(k)}\mathcal{V} = \eta J_N$, where $\eta[P] = [v_h]_h^{\otimes k} P(A_k)1$. Each local factor of $[v_h]_h$ has nonzero constant coefficient, so its inverse

is the finite geometric expansion in its nilpotent part. The kernel of $P \mapsto P(A_k)1$ is exactly (χ) by its definition. These two facts prove that η is injective, retaining every local derivative and arithmetic unit.

2 Return, action, and the entire original relation correction

Let $L_{i,j} : \mathcal{P}_i \hookrightarrow \mathcal{P}_j$ pad monic coefficients, and let $P_{i,j}$ be the orthogonal projection retaining their first $i+1$ entries. Then $B_j L_{i,j} = B_i$ and $L_{i,j}^* O_j L_{i,j} = O_i$. The original identity transport $I_{i,j} : (E, G_i) \rightarrow (E, G_j)$, $x \mapsto x$, is a contraction and intertwines A . Its metric adjoint is

$$T_{i,j} = K_i G_j : (E, G_j) \longrightarrow (E, G_i), \quad L_{i,j} R_i T_{i,j} = P_{i,j} R_j, \quad T_{i,j} T_{j,l} = T_{i,l}. \quad (\text{ERJ.4})$$

Indeed $x^* G_j y = x^* G_i T_{i,j} y$. The first $i+1$ coordinates of R_j are $O_i^{-1} B_i^* G_j$, exactly those of $R_i T_{i,j}$. The composition identity is $K_i G_j K_j G_l = K_i G_l$. For endomorphism types put $U_{i,j} = T_{i,j} I_{i,j}$ on (E, G_i) and $\widehat{U}_{i,j} = I_{i,j} T_{i,j}$ on (E, G_j) . Their coordinate matrices are $K_i G_j$, and the original invertible identity transport satisfies $I_{i,j} U_{i,j} = \widehat{U}_{i,j} I_{i,j}$. An endomorphism power or eigenvalue of T below means these specified similar endomorphisms, with all metrics retained. As an underlying endomorphism of E , $T = T_{i,j}$ is self-adjoint and positive in both displayed metrics: $G_i T = G_j$ and $G_j T = G_j K_i G_j$. It obeys $0 < T \preceq I$ in each metric. Its eigenvalues g_a lie in $(0, 1]$. The squared singular values of $I_{i,j}$, of its adjoint in the reverse Hilbert types, and of the restricted canonical lift in (ERJ.4) are these same g_a . Consequently

$$H_{i,j}^{\text{loss}} = I - T_{i,j}, \quad \mathcal{L}_{i,j} := \log(V_i/V_j) = -\log \det T_{i,j} = -\sum_{a=1}^q \log g_a. \quad (\text{ERJ.5})$$

Write $W_N = A^* G_N + G_N A - k G_N$, $H_N = K_N W_N$, and $V_N = A K_N + K_N A^* - k K_N$. Direct expansion proves

$$AT - TA = H_i T - T H_j = V_i G_j - K_i W_j. \quad (\text{ERJ.6})$$

The two $K_i A^* G_j$ terms and the two $k K_i G_j$ terms cancel. If $\epsilon_N = \|H_N\|_{G_N}$, contraction of the adjoint gives $\|AT - TA\|_{(E, G_j) \rightarrow (E, G_i)} \leq \epsilon_i + \epsilon_j$. Thus the original action accompanies the return through its exact commutator, including on any chosen return eigenspace. The representative change is the entire map

$$\begin{aligned} \Delta_{i,j} &= L_{i,j} R_i - R_j = L_{i,j} R_i H_{i,j}^{\text{loss}} - (I - P_{i,j}) R_j, \\ B_j \Delta_{i,j} &= 0, \quad \text{im } \Delta_{i,j} \subseteq \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \Delta_{i,j}^* O_j \Delta_{i,j} = G_i - G_j. \end{aligned} \quad (\text{ERJ.7})$$

Both terms in the difference have remainder $H_{i,j}^{\text{loss}}$. Orthogonality to $\mathcal{D}_i$ follows for both complete lifts. Since R_j is orthogonal to $\mathcal{D}_j$, the equality $L_{i,j} R_i = R_j + \Delta_{i,j}$ is an orthogonal sum and proves the Gram.

For the explicit theta primitive keep $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees zero and one, with F_h of degree one and ϕ_0 of degree zero. The original compatible chain action D commutes with this differential. Successive monic division gives $\chi(\sum s_a) = \sum_a h(s_a) Q_a(\mathbf{s})$ in the tensor ideal. For any relation $\chi\kappa$ the cochain

$$\sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa(\sum_b D_b) (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}) \quad (\text{ERJ.8})$$

has differential $\mathcal{V}(\chi\kappa)$: applying the tensor differential gives a second factor $(-1)^{a-1}$ and $h(D_a) F_h = \Theta \phi_0$; all other terms have zero differential. Apply this fixed linear division primitive to every column of (ERJ.7). On an active split fibre the maps are $x^\bullet \mapsto x^\bullet$ and $x^\bullet \mapsto (Tx)^\bullet$, with $\tau \mapsto \tau$. They are $G(\mathbb{C})$ -linear by the original fibre operations and linearity of the displayed maps. The relation image in the internal quotient is the supported zero $(\ell_j, 0)$; the external τ is retained. The reverse maps compose by (ERJ.4), while their action comparison is (ERJ.6). No relation, support label, or primitive is deleted.

3 The exact source projection and principal-angle morphism

Use the common original Hilbert space $\mathcal{P}_j$ and its isometries $\ell_i = L_{i,j}R_i : (E, G_i) \rightarrow \mathcal{P}_j$ and $\ell_j = R_j : (E, G_j) \rightarrow \mathcal{P}_j$. Let Π_i, Π_j be the orthogonal endomorphism projections onto these two images. Because $R_j \perp \mathcal{D}_j \supseteq \mathcal{D}_i$, its lower-degree projection lies in $\ell_i E$. Also $\ell_i x - \ell_j x \in \mathcal{D}_j$. Hence

$$\Pi_i \ell_j = P_{i,j} \ell_j = \ell_i T, \quad \Pi_j \ell_i = \ell_j, \quad (\Pi_i \Pi_j)|_{\ell_i E} = \ell_i T \ell_i^{-1}, \quad (\Pi_j \Pi_i)|_{\ell_j E} = \ell_j T \ell_j^{-1}. \quad (\text{ERJ.9})$$

These are similarities on the indicated full images. The range of $\Pi_i \Pi_j$ is $\ell_i E$ because its restriction there is invertible. Its kernel is $\ker \Pi_j$: the reverse inclusion follows from invertibility of $\Pi_i : \ell_j E \rightarrow \ell_i E$. This gives its direct sum into that image and kernel, with exactly q nonzero eigenvalues g_a and $j + 1 - q$ zero eigenvalues. Reversing the pair proves the corresponding assertion for $\Pi_j \Pi_i$.

Choose a G_i -orthonormal eigenbasis x_a of T . Put $u_a = \ell_i x_a$ and $v_a = \ell_j x_a / \sqrt{g_a}$. Since $G_j = G_i T$, both families are orthonormal and $\langle u_a, v_b \rangle = \delta_{ab} \sqrt{g_a}$ by (ERJ.9). If $g_a = 1$ then $u_a = v_a$. Otherwise $w_a = (v_a - \sqrt{g_a} u_a) / \sqrt{1 - g_a}$ makes mutually orthonormal pairs (u_a, w_a) . On each such plane,

$$\Pi_i = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \Pi_j = \begin{pmatrix} g_a & \sqrt{g_a(1-g_a)} \\ \sqrt{g_a(1-g_a)} & 1-g_a \end{pmatrix}. \quad (\text{ERJ.10})$$

Their difference has trace zero and square $(1 - g_a)I_2$. Both vanish on the common orthogonal complement, and the shared directions with $g_a = 1$ contribute zero. Therefore, for every real $s > 0$ and in particular each integer $m \geq 1$,

$$\text{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2s} = 2 \sum_a (1 - g_a)^s, \quad \text{Tr}_{\mathcal{P}_j} |\Pi_i - \Pi_j|^{2m} = 2 \text{Tr}_E ((H_{i,j}^{\text{loss}})^m). \quad (\text{ERJ.11})$$

Functional powers here use the proved positive spectral decomposition. The restricted map $\mathcal{R} = \Pi_i|_{\ell_j E}$ is the bijection $\ell_i T_{i,j} \ell_j^{-1} : \ell_j E \rightarrow \ell_i E$. For every $s > 0$ the exact inverse singular-value-power observation is

$$\sum_a \sigma_a(\mathcal{R}^{-1})^{2s} = \sum_a \sigma_a(T_{i,j}^{-1})^{2s} = \sum_a \sigma_a(I_{i,j}^{-1})^{2s} = \text{Tr } U_{i,j}^{-s} = \sum_a g_a^{-s}. \quad (\text{ERJ.11a})$$

This follows from $I_{i,j}^\dagger I_{i,j} = U_{i,j}$ and the equal singular values of a map and its adjoint; inversion reciprocates the singular values. These are Schatten observations of the original typed maps, including when $0 < 2s < 1$, where no norm triangle inequality is asserted. Only the bijection between the canonical images is inverted; the full projection products retain their ambient zeros. For a return eigenvector x of eigenvalue g the original lower portion satisfies

$$\frac{\|P_{i,j} R_j x\|_{\mathcal{O}_j}^2}{\|R_j x\|_{\mathcal{O}_j}^2} = g, \quad \frac{\|(I - P_{i,j}) R_j x\|_{\mathcal{O}_j}^2}{\|R_j x\|_{\mathcal{O}_j}^2} = 1 - g. \quad (\text{ERJ.12})$$

Indeed its first squared norm is $x^* G_j K_i G_j x = g x^* G_j x$; orthogonal decomposition proves the second. Its old quotient coordinate is $g x$. Applying the invertible T^{-1} recovers x ; the complete representative difference is exactly (ERJ.7).

4 The arithmetic remainder bound at every finite degree

For the actual quartet $1/2 \pm \delta \pm i\gamma$ of common order $m \geq 1$, $\delta, \gamma > 0$, retain

$$\ell = 1 + k(m - 1), \quad q = \ell(k + 1)^2, \quad \chi(S) = \prod_{a,b=0}^k (S - c - (2a - k)\delta - i(2b - k)\gamma)^\ell. \quad (\text{ERJ.13})$$

The top nonzero power of the sum of k local length- m nilpotents is $k(m-1)$, with coefficient $(k(m-1))!/((m-1)!)^k$ on the product of their top powers; its next power vanishes by total degree. Each displayed grid sum occurs and the distinct pairs have distinct real and imaginary parts. This proves the complete order and degree in (ERJ.13) by taking the least common multiple of local annihilators.

The complete arithmetic-envelope source proves, for $k \geq 3$,

$$m_k(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h, \quad \vartheta_h = \int_{-1}^1 w_h > 0. \quad (\text{ERJ.14})$$

Here $c_h > 0$ is its original local-mass convolution constant. Fix $0 < b < \pi/2$, $M_h(\pm b) = \int e^{\pm bt} w_h(t) dt$ and $M_k = \mu_h^k + M_k(b)^k + M_k(-b)^k$. Positivity and convolution give $\int e^{b|u|} m_k \leq M_k$, and hence $\int |u|^{2d} m_k \leq M_k (2d)! b^{-2d}$. Set $R_0 = \sqrt{\delta^2 + \gamma^2}$, $D = \max(1, 2/b, 2R_0)$, $T = Dq$, $r_0 = kR_0/T \leq 1/2$, and retain

$$\begin{aligned} a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(k-3+T)} (k-2+T)^{-B}, \\ U_k(T) &= \mu_h^k + M_k \frac{(2q-2)!}{(bT)^{2q-2}}, \quad A_N = \sqrt{(2N+1)/2} \frac{4^{N+1} - 1}{3}, \\ B_q(r_0) &= \left(\frac{1+r_0}{1-r_0} \right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r_0)^2}{T a_k(T)}. \end{aligned} \quad (\text{ERJ.15})$$

The coordinate map $\Psi_T P(x) = P(c + iTx)$ has inverse $f \mapsto f((S-c)/(iT))$ and target measure $T m_k(Tx) dx$, whose mass remains μ_h^k . Put $\chi_T = (iT)^{-q} \chi(c + iTx)$. Then $\Psi_T(\chi Q) = (iT)^q \chi_T \Psi_T Q$, so remainder commutes with this map. Every original raw derivative of order d is multiplied by $(iT)^d$. The quotient coefficient determinant is $(iT)^{q(q-1)/2}$, and its transported form determinant is $T^{-q(q-1)} V_N$; this identical factor cancels in a volume ratio.

List all q roots z_a of χ_T with repetition. Their moduli are at most r_0 . Write $E(t) = \prod_a (1 - z_a t)$ and $E(t)^{-1} = \sum_{l \geq 0} h_l t^l$. For every integer $s \geq 0$, multiplication of these formal inverse series through order s proves that $Q_s(x) = \sum_{l=0}^s h_l x^{s-l}$ is the quotient of x^{q+s} by χ_T , and that $x^{q+s} - \chi_T Q_s$ has degree below q . Convolution gives $\|\chi_T\|_1 \leq (1+r_0)^q$ and $|h_l| \leq \binom{q+l-1}{l} r_0^l$, where $\|\cdot\|_1$ is the coefficient absolute sum. The sole coefficient of $\chi_T Q_s$ at degree at least q is its leading coefficient one. Its remaining coefficients are the negative remainder. Therefore

$$\|\text{rem}_{\chi_T} x^{q+s}\|_1 \leq (1+r_0)^q \sum_{l=0}^s \binom{q+l-1}{l} r_0^l - 1 \leq B_q(r_0). \quad (\text{ERJ.16})$$

The last bound sums the q geometric series. At degrees below q the remainder has norm one. Linearity proves $\|\text{rem}_{\chi_T} f\|_1 \leq B_q(r_0) \|f\|_1$ for every polynomial f , without a degree restriction on s .

Rodrigues' formula for Legendre polynomials gives orthogonality, squared norm $2/(2n+1)$, and coefficient norm at most $\binom{2n}{n} \leq 4^n$. For the norm, integrate by parts n times; the endpoint terms vanish because $(x^2-1)^n$ has order n there, and the remaining beta integral gives $2/(2n+1)$. For the coefficient bound use $P_n(x) = 2^{-n} \sum_{a=0}^n \binom{n}{a}^2 (x-1)^{n-a} (x+1)^a$, then sum the coefficient absolute values. Cauchy-Schwarz on each orthogonal coefficient and summation of 4^n yield $\|f\|_1 \leq A_N \|f\|_{L^2[-1,1]}$ for $\deg f \leq N$. For $\deg v \leq q-1$, each monomial at u/T has modulus at most $\max(1, |u/T|^{q-1})$, so $\int |v(u/T)|^2 m_k \leq U_k(T) \|v\|_1^2$. Finally (ERJ.14) restricted to $[-T, T]$ gives $\|P\|^2 \geq T a_k(T) \|\Psi_T P\|_{L^2[-1,1]}^2$. Composing these actual maps proves, for every $N \geq q-1$,

$$\|\text{rem}_\chi P\|^2 \leq C_N \|P\|^2, \quad G_{q-1} \preceq C_N G_N, \quad C_j^{-1} K_j \preceq K_i \preceq K_j \quad (q-1 \leq i \leq j). \quad (\text{ERJ.17})$$

Apply the first inequality to the least lift $R_N x$ to obtain the second. Then use $G_j \preceq G_i \preceq G_{q-1}$ and inversion of positive forms. Taking $i = j$ proves $C_j \geq 1$. Thus the two original endpoint pairs $(q-1, 2q-1)$ and $(q, 2q)$ have the explicit positive gaps C_{2q-1}^{-1} and C_{2q}^{-1} .

5 Finite comparison matrices and the complete trace error

All following comparisons use the original quotient coordinates. For positive source brackets $H_N^- \preceq H_N \preceq H_N^+$, inversion and congruence give

$$L_N = J_N(H_N^+)^{-1} J_N^* \preceq K_N \preceq U_N = J_N(H_N^-)^{-1} J_N^*. \quad (\text{ERJ.18})$$

Their positivity follows from surjectivity of J_N . One actual upper comparator uses the density $\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3+|u|)} (k-2+|u|)^{-B}$ from (ERJ.14). Its even moment of degree d is

$$\int u^d \nu_k(u) du = \frac{2c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} d!}{(B-1)!} \int_0^\infty \frac{t^{B-1} e^{-(k-2)t}}{(\alpha+t)^{d+1}} dt; \quad \int u^d \nu_k(u) du = 0 \quad (d \text{ odd}). \quad (\text{ERJ.19})$$

Insert $(k-2+u)^{-B} = (B-1)!^{-1} \int_0^\infty t^{B-1} e^{-(k-2+u)t} dt$ on the positive half-line; Tonelli and $\int_0^\infty u^d e^{-yu} du = d! y^{-d-1}$ prove the formula. Thus $H_N^\nu \preceq H_N$ is an explicit strictly positive source. It supplies U_j using comparison moments through $2j$, at most $4q$ at the second endpoint; actual low-degree source data for i stop at $2i \leq 2q$. These cutoffs keep the actual and comparison integrals attached to their stated maps.

Given $0 < L_i \preceq K_i \preceq K_j \preceq U_j$, set $\tau = \text{Tr}(L_i^{-1} U_j)$. The Hermitian matrix $L_i^{-1/2} U_j L_i^{-1/2}$ has all q eigenvalues at least one. Each is at most $\tau - q + 1$ because the remaining $q-1$ contribute at least $q-1$. It follows that

$$U_j \preceq (\tau - q + 1) L_i, \quad g := \frac{1}{\tau - q + 1} > 0, \quad K_i \succeq g K_j. \quad (\text{ERJ.20})$$

This coefficient is optimal from the trace and the lower bound one: $\text{diag}(\tau - q + 1, 1, \dots, 1)$ attains it. An upper rational $\tau_+ \geq \tau$ gives the certified rational gap $(\tau_+ - q + 1)^{-1}$.

Now take rational Hermitian brackets for both K_i, K_j as in (ERJ.18), and define the exact rational quantities

$$H_* = I - L_i U_j^{-1}, \quad t_m^* = \text{Tr } H_*^m, \quad d = \text{Tr}(U_i L_j^{-1}) - \text{Tr}(L_i U_j^{-1}), \quad r = 1 - g. \quad (\text{ERJ.21})$$

Here g is certified for the reference pair (L_i, U_j) itself. For increasing generalized eigenvalues the exact minimum-maximum formula is

$$\lambda_a(F, G) = \min_{\dim W = a} \max_{0 \neq x \in W} \frac{x^* F x}{x^* G x}. \quad (\text{ERJ.22})$$

For $G = I$, the span of the first a eigenvectors attains λ_a ; every a -dimensional W intersects the span of eigenvectors $a, \dots, q$, whose dimension is $q - a + 1$, and hence its maximum is at least λ_a . The invertible substitution $y = G^{1/2} x$ proves the general case with subspace dimensions preserved. The pointwise Rayleigh quotients for (L_i, U_j) , (K_i, K_j) , and (U_i, L_j) are ordered. Their extrema are therefore ordered. If $\alpha_a = \lambda_a(L_i, U_j)$ and $\beta_a = \lambda_a(K_i, K_j)$, this proves

$$g \leq \alpha_a \leq \beta_a \leq 1, \quad 0 \leq \sum_a (\beta_a - \alpha_a) \leq d. \quad (\text{ERJ.23})$$

The sum identity uses similarity of FG^{-1} to its positive Hermitian representative. The kernels F, G are forms on the dual quotient coefficient space: $G^{-1}F$ there and FG^{-1} on the quotient are similar through the invertible map G . These β_a are the actual g_a of the return. For $r > 0$, integration of the derivative of x^m on $[0, r]$ gives the complete moment enclosure

$$a_m := \max(0, t_m^* - m r^{m-1} d) \leq s_m := \text{Tr}((H_{i,j}^{\text{loss}})^m) \leq t_m^* \leq q r^m. \quad (\text{ERJ.24})$$

Indeed apply $0 \leq u^m - v^m \leq m r^{m-1}(u - v)$ to $u = 1 - \alpha_a$, $v = 1 - \beta_a$ and sum (ERJ.23). If $r = 0$, all $\alpha_a = \beta_a = 1$ and both loss operators vanish; the exact certificate is zero.

Integrating the geometric series on $[0, x]$ proves $-\log(1-x) = \sum_{m \geq 1} x^m/m$ for $0 \leq x < 1$. All terms are nonnegative. Its tail is bounded by $x^{p+1}/((p+1)(1-x))$. Thus for every integer $p \geq 1$,

$$\sum_{m=1}^p \frac{a_m}{m} \leq \mathcal{L}_{i,j} \leq \sum_{m=1}^p \frac{t_m^*}{m} + \frac{t_{p+1}^*}{(p+1)g}. \quad (\text{ERJ.25})$$

The width of the two stated endpoints is at most

$$\frac{d(1-r^p)}{g} + \frac{qr^{p+1}}{(p+1)g}. \quad (\text{ERJ.26})$$

The first term sums $r^{m-1}d$ using (ERJ.24); the second uses $t_{p+1}^* \leq qr^{p+1}$. This accounts for the finite matrix error and series tail separately with their original dimension.

The sharper coefficient

$$c_p(r) = \sum_{l \geq 0} \frac{r^l}{p+1+l} = \frac{-\log g - \sum_{m=1}^p r^m/m}{r^{p+1}} \leq \frac{1}{(p+1)g} \quad (\text{ERJ.27})$$

can replace the last coefficient in (ERJ.25): the tail at $x \leq r$ is at most $c_p(r)x^{p+1}$. To certify its logarithm, write exactly $x = 2^e y$ for rational $x > 0$, integer e , and $1 \leq y < 2$. For $t = (y-1)/(y+1)$, integration of $2/(1-t^2)$ gives

$$\log y = 2 \sum_{a=0}^{M-1} \frac{t^{2a+1}}{2a+1} + E_M, \quad 0 \leq E_M \leq \frac{2t^{2M+1}}{(2M+1)(1-t^2)}. \quad (\text{ERJ.28})$$

At $t = 1/3$ the same formula encloses $\log 2$. Multiply that interval by e , reversing endpoints when $e < 0$, and add the $\log y$ interval. If $\log g \in [\ell_-, \ell_+]$ has width ε , replacing $-\log g$ by $-\ell_-$ in (ERJ.27) adds at most $\varepsilon t_{p+1}^*/r^{p+1} \leq q\varepsilon$ to the upper endpoint. The $r = 0$ case was handled before this division.

6 Constructing rational brackets for the actual relation

Let an actual Hermitian source matrix M have a rational Hermitian center M_0 and symmetric rational radii $\varepsilon_{ab} \geq 0$ with $|(M - M_0)_{ab}| \leq \varepsilon_{ab}$. For fixed positive rational w_a set $D_{aa} = \sum_b \varepsilon_{ab} w_b / w_a$. Squaring $\sqrt{t}x - y/\sqrt{t}$ proves $2xy \leq tx^2 + t^{-1}y^2$. Therefore

$$\begin{aligned} |z^*(M - M_0)z| &\leq \sum_{a,b} \varepsilon_{ab} |z_a| |z_b| \\ &\leq \frac{1}{2} \sum_{a,b} \varepsilon_{ab} \left(\frac{w_b}{w_a} |z_a|^2 + \frac{w_a}{w_b} |z_b|^2 \right) = z^* D z. \end{aligned} \quad (\text{ERJ.29})$$

Hence $M_0 - D \preceq M \preceq M_0 + D$. Exact Hermitian elimination tests positivity: a first pivot $d_0 > 0$ gives $d_0|x + d_0^{-1}b^*y|^2 + y^*(C - bb^*/d_0)y$. Induction proves positivity precisely when all successive pivots and their Schur complements are positive.

For the actual, possibly nonrational coefficients of χ , use the exact ordered coefficient matrix

$$\mathcal{A}_N = [1, S, \dots, S^{q-1}, \chi, S\chi, \dots, S^{N-q}\chi]. \quad (\text{ERJ.30})$$

It is triangular with diagonal one. At $N = q - 1$ its relation block is empty. Monic division gives $J_N \mathcal{A}_N = [I_q, 0] =: J_0$. Consequently, in the original remainder coordinates,

$$M_N = \mathcal{A}_N^* H_N \mathcal{A}_N, \quad K_N = J_0 M_N^{-1} J_0^*. \quad (\text{ERJ.31})$$

Intervals for the original source entries and original coefficients of χ enter the literal finite products for M_N ; interval addition, multiplication, and conjugation contain those products. The sum of the real and imaginary radii bounds a complex error. Use conjugate centers and symmetrically enlarged radii, then (ERJ.29) and exact positive pivots. The two inverse compressions of the resulting source brackets yield rational L_N, U_N by (ERJ.18). No center polynomial replaces the actual quotient map in (ERJ.31).

For a fixed actual pair and fixed w_a , refining source and coefficient intervals makes the radii and D tend to zero. Positivity of the actual M_N ensures eventual positive lower pivots. The cofactor formula proves continuity of inversion where the determinant is nonzero; thus $L_N, U_N \rightarrow K_N$, $d \rightarrow 0$, and the reference gap in (ERJ.20) tends to a positive number. For any requested positive width, first make d/g sufficiently small and then choose p so the geometric tail in (ERJ.26) is smaller than the remaining width. The rational logarithm error can be decreased independently. This proves convergence of the finite algebraic certificate for each fixed actual pair. The quantities controlling that convergence remain explicit when the degree is allowed to grow.

7 The four-volume comparison with its literal finite constants

The arithmetic pair losses add to $\mathcal{B}_{h,k} = \log(V_{q-1}V_q/(V_{2q-1}V_{2q}))$. Equations (ERJ.15)–(ERJ.17) prove

$$\mathcal{B}_{h,k} \leq q \log C_{2q-1} + q \log C_{2q}, \quad \frac{C_{2q}}{C_{2q-1}} = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1}-1}{4^{2q}-1} \right)^2 \leq \frac{147}{5} < 32. \quad (\text{ERJ.32})$$

Indeed the first ratio is at most $5/3$, and the ratio inside the square is $4 + 3/(4^{2q}-1) \leq 21/5$. Set $X_0 = \max(1, M_h(b), M_h(-b))$, $K_0 = \max(1, \vartheta_h^{-1})$ and $C_{\text{env}} = \max(1, 2/(3c_h D))$. Since $bT \geq 2q$, the factorial ratio in U_k is at most one; also $M_k = 2 \int \cosh(bu) m_k \geq 2\mu_h^k$. Thus $U_k(T) \leq 3X_0^k$. The stated coefficient constant obeys $A_{2q-1}^2 \leq (2q/9)256^q$. On $0 \leq r_0 \leq 1/2$, the derivative of $\log((1+r_0)/(1-r_0))$ is at most $8/3$, so $\log B_q(r_0)^2 \leq 16R_0 k/(3D)$. Substitution into C_{2q-1} , using $\vartheta_h^{-(k-3)} \leq K_0^k$ and $k-2+Dq \leq 1+(D+1)q$, proves

$$q \log C_{2q-1} \leq \mathcal{U}_k, \quad \mathcal{B}_{h,k} \leq 2\mathcal{U}_k + q \log 32, \\ \mathcal{U}_k = (\alpha D + \log 256)q^2 + \left(\log(X_0 K_0) + \alpha + \frac{16R_0}{3D} \right) kq + Bq \log(1 + (D+1)q) + q \log C_{\text{env}}. \quad (\text{ERJ.33})$$

All constants are independent of k for the fixed packet and b . As $q = [1 + k(m-1)](k+1)^2$, $k/q \rightarrow 0$ and $\log q/q \rightarrow 0$. The displayed scalar majorant divided by q^2 tends to $2(\alpha D + \log 256)$; divided by $q \log k$ it grows. The finite matrix and trace bounds (ERJ.18)–(ERJ.28) observe the same original four volumes and retain the directional data before this scalar estimate.

8 Central rank, both endpoint returns, and all inverse powers

Keep the original phase ϕ_N and relative Hermitian radius ϵ_N . For a reflection-stable packet the exact monic recurrence is $Sp_n = p_{n+1} + (c + ia_n)p_n - (\omega_n/\omega_{n-1})p_{n-1}$, $a_n \in \mathbb{R}$. It follows by transporting to u with $p_n(c + iu) = i^n Q_n(u)$ and pairing self-adjoint multiplication by u with every lower monic orthogonal polynomial. Substitution into (ERJ.3) cancels the internal adjacent terms and gives

$$\mathbf{V}_N = (b_{N+1}b_N^* + b_N b_{N+1}^*)/\omega_N, \quad b_N^* G_N b_{N+1} = i\omega_N \phi_N. \quad (\text{ERJ.34})$$

The pairing in the second formula is pure imaginary: in the real u coordinates both the quotient Gram and the remainder columns are real, and the source factors contribute $i^{\overline{N}} i^{N+1} = i$. This defines the retained real phase; it equals the original phase in the monic norm derivative formula. The relative Hermitian matrix has rank at most two and trace zero. Its square has trace

$2[(b_N^* G_N b_N)(b_{N+1}^* G_N b_{N+1}) - |b_N^* G_N b_{N+1}|^2]/\omega_N^2$, giving eigenvalues $\epsilon_N, -\epsilon_N$ and the remaining zeros. The rank-one determinant update in (ERJ.3) gives, for $N \geq q$, $b_N^* G_N b_N/\omega_N = 1 - \delta_N$ and $b_{N+1}^* G_N b_{N+1}/\omega_{N+1} = \delta_{N+1}^{-1} - 1$, where $\delta_N = V_N/V_{N-1}$. Therefore

$$\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N}(1 - \delta_N)(\delta_{N+1}^{-1} - 1). \quad (\text{ERJ.35})$$

The determinant update follows by eliminating $\begin{pmatrix} I_Y & X \\ -Y & I \end{pmatrix}$ in its two block orders to prove $\det(I + XY) = \det(I + YX)$. Multiplying (ERJ.35) for $N = q, \dots, 2q - 2$ proves

$$\prod_{N=q}^{2q-2} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_{2q-1} V_q}{\omega_q V_{2q-1}} (1 - d_0)(1 - d_{q-1}) \prod_{a=1}^{q-2} (1 - d_a)^2, \quad d_a = \delta_{q+a}. \quad (\text{ERJ.36})$$

Each internal contraction occurs twice and both endpoint contractions occur once.

For (ERJ.13) the complete positive-real spectral compression gives

$$L = 2\delta\ell(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad L/q \geq \delta k/2, \quad 0 < L \leq \epsilon_N. \quad (\text{ERJ.37})$$

Here twice the sum of positive real parts of the eigenvalues of $A - cI$ equals L : sum the positive integers $2a - k$, giving $\lfloor (k+1)^2/4 \rfloor$, then include all $k+1$ imaginary centres and their full length ℓ . A unitary upper triangular form in the actual G_N metric, with these eigenvalues first, shows that this is the trace of the Hermitian defect compressed to their invariant subspace. A projection paired with a Hermitian matrix of eigenvalues $\epsilon_N, -\epsilon_N, 0, \dots, 0$ has trace at most ϵ_N , proving the stated bound. The integer inequality $\lfloor (k+1)^2/4 \rfloor \geq k(k+1)/4$ follows in each parity.

The complete balanced monic-norm source proves

$$\begin{aligned} (\omega_{n+r}/\omega_n)^{1/(2r)} &\leq C_h^{\text{bal}} n \quad (n \geq k \geq 3, \quad n/2 \leq r \leq n), \\ C_h^{\text{bal}} &= 256e^\pi \max(1, 3/c_h) X_0 K_0 \max(1, b^{-3}) 6^{B/3}. \end{aligned} \quad (\text{ERJ.38})$$

This is its literal sufficient constant from the original envelope and monic Legendre minimum, with its full analytic proof included in the dependency record. Apply it at $n = q, r = q - 1$ and put $X = L/(C_h^{\text{bal}} q)$. Every d_a in (ERJ.36) is strictly below one, since otherwise its right side is zero while every left factor is at least $L^2 > 0$. Define the entire nonnegative penalty

$$\begin{aligned} P_{\text{mid}} &= -\log(1 - d_0) - \log(1 - d_{q-1}) - 2 \sum_{a=1}^{q-2} \log(1 - d_a) \\ &\quad + \sum_{a=0}^{q-2} \log(1 + \phi_{q+a}^2/L^2). \end{aligned} \quad (\text{ERJ.39})$$

Taking logarithms in (ERJ.36) proves

$$C_{\text{mid}} := \log(V_q/V_{2q-1}) \geq 2(q-1) \log X + P_{\text{mid}}. \quad (\text{ERJ.40})$$

Indeed every factor $\epsilon_N^2 + \phi_N^2$ is at least $L^2(1 + \phi_N^2/L^2)$, and (ERJ.38) bounds the norm ratio.

The central update is the full matrix

$$K_{2q-1} - K_q = \sum_{a=q+1}^{2q-1} b_a b_a^*/\omega_a, \quad \text{rank}(K_{2q-1} - K_q) \leq q - 1. \quad (\text{ERJ.41})$$

Its positive Hermitian representative shows that the central return has at least one exact unit eigenvalue. List one such value separately and the other $q - 1$ as $\gamma_1, \dots, \gamma_{q-1}$, keeping any

additional unit values. Their product is $e^{-C_{\text{mid}}}$. For every real $s > 0$, convexity of e^{sx} gives

$$\begin{aligned} \text{Tr } T_{q,2q-1}^{-s} &= 1 + \sum_{a=1}^{q-1} \gamma_a^{-s} \\ &\geq 1 + (q-1)e^{sC_{\text{mid}}/(q-1)} \\ &\geq \max \left\{ q, 1 + (q-1)X^{2s}e^{sP_{\text{mid}}/(q-1)} \right\}. \end{aligned} \tag{ERJ.42}$$

For completeness $e^y \geq 1 + y$ follows by differentiating $e^y - 1 - y$, whose minimum is zero at $y = 0$. Apply it to $y_a = -s \log \gamma_a - \bar{y}$, where $\bar{y} = -(s/(q-1)) \sum_a \log \gamma_a$: $\sum_a e^{-s \log \gamma_a} = e^{\bar{y}} \sum_a e^{y_a} \geq (q-1)e^{\bar{y}}$. The q bound also follows termwise from $\gamma_a \leq 1$. The actual rank $\rho = \text{rank}(K_{2q-1} - K_q)$ preserves exactly $q - \rho$ unit eigenvalues, because $I - T = (K_{2q-1} - K_q)G_{2q-1}$ has rank ρ . Write $A_+ = \max\{0, 2(q-1) \log X + P_{\text{mid}}\}$. If $\rho > 0$, applying the same exponential argument to just the ρ eigenvalues below one proves the stronger exact-rank consequence

$$\text{Tr } T_{q,2q-1}^{-s} \geq q - \rho + \rho e^{sC_{\text{mid}}/\rho} \geq q - \rho + \rho e^{sA_+/\rho}. \tag{ERJ.42a}$$

For fixed $a \geq 0$, $r(e^{a/r} - 1)$ decreases as $r > 0$ increases: its derivative is $e^{a/r}(1 - a/r) - 1 \leq 0$, since the derivative of $e^t(1 - t)$ is $-te^t \leq 0$ for $t \geq 0$. Thus $\rho \leq q - 1$ proves that (ERJ.42a) improves (ERJ.42). At $\rho = 0$ the central return is the identity, its cost is zero, and the trace is exactly q . There is no division by ρ in that case. By (ERJ.22), the ordered eigenvalues of each pair (K_{q-1}, K_{2q-1}) and (K_q, K_{2q}) are at most those of (K_q, K_{2q-1}) : their Rayleigh quotients decrease respectively in the numerator and by increase of the denominator. Since x^{-s} is decreasing, *each* original endpoint return obeys the final bound in (ERJ.42), and the stronger rank-refined bound (ERJ.42a), by the same ordered comparison. Its minimum eigenvalue also obeys

$$g_{\min}^{(a)} \leq \min \left\{ 1, X^{-2}e^{-P_{\text{mid}}/(q-1)} \right\} \leq \min \left\{ 1, \frac{4(C_h^{\text{bal}})^2}{\delta^2 k^2} \right\}, \quad a = 0, 1. \tag{ERJ.43}$$

The first inequality follows from $\gamma_{\min}^{q-1} \leq \prod \gamma_a$ and (ERJ.40), followed by the same ordered comparison. Equations (ERJ.9)–(ERJ.12) and (ERJ.7)–(ERJ.8) give its exact source, action, and theta interpretation. The identity

$$\mathcal{B}_{h,k} = 2C_{\text{mid}} + \log(V_{q-1}/V_q) + \log(V_{2q-1}/V_{2q}) \tag{ERJ.44}$$

has nonnegative endpoint terms. Combining (ERJ.37)–(ERJ.40) proves

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{B}_{h,k}}{q_k \log k} \geq 4,$$

because its lower bound is $4(q-1)[\log k + \log(\delta/(2C_h^{\text{bal}}))]$ and $(q-1)/q \rightarrow 1$. The consecutive-first-window continuation separately proves the length- q products at the two actual endpoints, including the single terminal factor at initial degree $q - 1$, and gives its stronger per-window inverse-power estimate. Its original constant and phase formula stay in that complete companion proof; (ERJ.42) retains the exact central unit eigenvalue and full central penalty.

9 Exact auxiliary calculation and retained proof dependencies

The accompanying complete fixture proof uses $S = 1 + iu$, $\chi = S^2 - 2S - 1$ and $r_\lambda(u) = |\Gamma(\lambda + iu/2)|^2/(2\pi)$. Its source is $m = r_4/70$ of mass $9/16$; its smaller source is $\nu = (54/35)(r_2/3)$ of mass $27/70$. The full identity

$$m(u) = \left(\frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120} \right) \frac{r_2(u)}{3} \tag{ERJ.45}$$

follows from two applications of $\Gamma(z + 1) = z\Gamma(z)$, and proves their source contraction. The complete fixture appendix derives all moments, original S Grams, remainder matrices, extrema, and rational logarithm endpoints. Its remainder norm squared is $763/108$, attained by the original polynomial $-245/36 + (791/108)S - (7/9)S^2 + (7/27)S^3$ in both degree three and degree four. The resulting source gap is $108/763$, while the two actual minimum return eigenvalues equal $135/331$. The exact four-volume ratio is $3073295611/216513000$. These quantities are proved in the full fixture text retained with this article, rather than inferred from a decimal computation.

The existing fixture executed 303 distinct mathematical checks in each unchanged normal and optimized run. Seven literal formula changes in each mode were rejected after the complete calculation, giving 16 actual jobs in total. The preserved source and full execution JSON have hashes recorded in this article's manifest. This completion did not rerun that unchanged suite. Its exact source/Gamma fixture is connected to the arithmetic construction by (ERJ.2)–(ERJ.4), (ERJ.17)–(ERJ.20); the density (ERJ.45) and its masses remain its declared auxiliary inputs.

For the empty packet $h = 1$, $E = 0$, the quotient determinants are empty determinants one and every quotient trace and four-volume loss is zero. The original analytic source and mass still exist. The positive inverses and $q - 1$ spectral averages above are used only in their stated domains; the actual quartet has $q \geq (k + 1)^2$ and $k \geq 3$. At $i = j$, return is the identity, its relation correction and projection difference vanish, and the exact loss is zero.

The complete source dependencies are the original *Canonical restriction and the arithmetic endpoint volume*; *An arithmetic lower-envelope comparison for central quotient volumes* (AU.1–49); *Exact window products, sharp endpoint optimization, and original relation costs* (EP.1–54), with its complete balanced-window proof dependency; the endpoint-window proof (EW.1–59); the complete 303-check fixture proof; and the *Consecutive first-window join* (CJ.1–22). The accompanying manifest pins their actual files and the new independent mathematical review. The global cumulative reader carries these full proofs and the new article in their stated order.